Puja Majumder

Assistant Professor | Principal Investigator, Majumder Lab

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Academic appointments

June 2026 - present | Assistant Professor
Department of Bioscience and Biotechnology, IIT Kharagpur, India

2025 - 2026 | Senior Research Scientist
Memorial Sloan Kettering Cancer Center, New York, USA

2024 - 2025 | Research Associate
Memorial Sloan Kettering Cancer Center, New York, USA

2022 - 2024 | Research Scholar
Memorial Sloan Kettering Cancer Center, New York, USA | Dinshaw J. Patel Lab

February 2021 - March 2022 | Research Associate
Molecular Biophysics Unit, Indian Institute of Science, Bengaluru, India

Education

2015 - 2021 | PhD in Structural Biology
Indian Institute of Science, Bengaluru
Supervisor: Prof. Aravind Penmatsa
Thesis: Structural and Functional Investigation of a Multi-drug Efflux Transporter QacA

2013 - 2015 | MSc in Biotechnology
Jawaharlal Nehru University, New Delhi
First Class First, Gold Medal

2011 - 2013 | BSc Honours in Microbiology
St. Xavier's College (Autonomous), Kolkata
First Class Honours

Awards and fellowships

2025 Ludwig Cancer Research Postdoctoral Fellowship and Award

2019 Best Poster Award, EMBO Workshop on Tools for Structural Biology of Membrane Proteins

2018 Best Poster Award, Indian Biophysical Society Meeting, IISER Pune

2015 GATE Doctoral Fellowship; DST INSPIRE Scholarship; MSc Gold Medal, JNU

2013 CEEB Fellowship, Combined Entrance Examination for Biotechnology

Research focus

I investigate how bacterial antiphage defense systems translate immune signals into molecular responses that restrict viral replication. My lab combines cryo-EM, X-ray crystallography, biophysics, biochemistry, microbiology, and computational biology to connect protein structure with immune function.

Selected research outputs

- **Cap1 | Accepted in Nature Structural & Molecular Biology (2026)**
  Co-first author
- **Chp1 | bioRxiv (2026)**
  First and co-corresponding author
- **Cat1 | Science (2025)**
  Co-first and co-corresponding author
- **Cad1 | Cell (2024)**
  Co-first author
- **Cam1 | Nature (2024)**
  Co-author
- **QacA | The EMBO Journal (2023)**
  First author

Scientific contributions

Membrane pores
Signal-triggered pore opening and lipid gating in CRISPR-associated effectors.

Metabolic changes
ATP-to-ITP conversion and filament-dependent NAD degradation during antiphage defense.

Antimicrobial efflux
Substrate recognition, proton coupling, and allosteric regulation in QacA.

Appointments and education

Academic appointments

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2024 - 2025 | **Research Associate**
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Methods and technical expertise

Cryo-EM Sample preparation, Vitrobot grid freezing and clipping; independent microscope use; EPU and SerialEM data collection; CryoSPARC and RELION processing; soluble and membrane proteins, filaments, and supramolecular complexes.

Crystallography and molecular analysis Protein crystallization, synchrotron data collection, structure determination and refinement; HKL-3000, CCP4, REFMAC, Coot, and Phenix.

Biochemistry and biophysics Protein purification; antibody, Fab, and nanobody approaches; ITC, MST, SEC-MALS, mass photometry, fluorescence spectroscopy, circular dichroism, nano-DSF, and membrane transport assays.

Computation and analysis GROMACS molecular dynamics; AlphaFold, Foldseek, DALI, and BLAST; ChimeraX, PyMOL, GraphPad Prism, Origin, and Illustrator.

Laboratory development and mentorship

Established the Majumder Lab at IIT Kharagpur in June 2026, focusing on bacterial antiphage defence through structural biology, biophysics, biochemistry, and microbiology. Bishnupada Mandal joined the PhD programme in November 2026 as the lab's first research scholar.

Contributions to science

CRISPR membrane pores and lipid gating

Membrane-associated immune effectors link intracellular infection signals to growth arrest, but the structural basis of their activation requires direct mechanistic analysis. Cap1 and Chp1 reveal distinct routes from cyclic oligoadenylate recognition to membrane depolarization. Cap1 opens a preformed tetrameric pore following cyclic tetra-adenylate binding. Chp1 undergoes a lipid-gated conformational switch that removes pore obstruction and permits ion flow. These studies connect second-messenger signalling to antiviral growth arrest. Cap1: co-first author; accepted in Nature Structural & Molecular Biology. Chp1: first and co-corresponding author, bioRxiv 2026.

Filament assembly and NAD depletion

TIR-domain immune proteins can couple higher-order assembly to catalytic activity. I resolved the active supramolecular assembly of the Cat1 CARF-TIR effector by cryo-EM. Structural and functional studies linked cyclic signal recognition, filament-network formation, and a composite NADase active site to depletion of cellular NAD during antiviral defence. Co-first and co-corresponding author, Science 2025.

Metabolic reprogramming in antiviral defence

Type III CRISPR-Cas effectors can defend cells by changing essential metabolite pools. I combined X-ray crystallography, cryo-EM, and biochemical characterization to define signal recognition and catalytic regulation in Cad1. The work established how a cyclic nucleotide-activated adenosine deaminase converts ATP to ITP, arrests infected-cell growth, and limits phage propagation. Co-first author, Cell 2024.

Antibody-assisted structural biology of membrane effectors

I used high-affinity antibody fragments as fiducial markers to resolve the small membrane-associated CRISPR effector Cam1. Its tetrameric architecture provides a structural basis for understanding membrane depolarization during type III CRISPR immunity. Nature 2024.

Antibacterial transport and allosteric regulation

Multidrug efflux pumps protect bacteria from chemically diverse antibacterial compounds and contribute to antimicrobial resistance. I determined the nanobody-bound QacA transporter structure at 3.6 angstrom resolution and used structure-guided functional assays to establish the role of an extracellular helical hairpin in efflux. Earlier work identified acidic residues governing substrate recognition, proton coupling, and multidrug transport. First-author papers in The EMBO Journal 2023 and Journal of Molecular Biology 2019.

Publications

Eight published journal articles and two preprints (one accepted for journal publication). Authorship contributions are indicated explicitly.

Preprints

1. **P. Majumder**[*], H. Carion, G. Stella, D. J. Patel[*], L. A. Marraffini[*]; A lipid fenestration-gated membrane depolarization mechanism expands the repertoire of CRISPR-mediated anti-phage defense strategies; bioRxiv, 2026.08.05.743128 (2026) · Chp1 preprint ([*]co-corresponding authors).

2. **P. Majumder**[‡], C. W. Cahir[‡], C. G. Roberts, D. J. Patel, L. A. Marraffini; Cap1 forms a cyclic tetra-adenylate-induced membrane pore during the type III-A CRISPR-Cas immune response; bioRxiv, 2025.11.13.688252 (2025) ([‡]co-first authors). Accepted in *Nature Structural & Molecular Biology*.

Journal articles

1. C. F. Baca[‡], **P. Majumder**[‡,*], J. H. Hickling, D. J. Patel, L. A. Marraffini[*]; Cat1 forms filament networks to degrade NAD+ during the type III CRISPR-Cas antiviral response; Science, 388, eadv9045 (2025) ([‡]co-first and *co-corresponding author)

2. C. F. Baca[‡], **P. Majumder**[‡], J. H. Hickling, L. Ye, M. Teplova, S. F. Brady, D. J. Patel, L. A. Marraffini; The CRISPR-associated adenosine deaminase Cad1 converts ATP to ITP to provide antiviral immunity; Cell, 187, 7183-7195.e24 (2024) ([‡]co-first author)

3. C. F. Baca, Y. Yu, J. T. Rostøl, **P. Majumder**, D. J. Patel, L. A. Marraffini; The CRISPR effector Cam1 mediates membrane depolarization for phage defence; Nature, 625, 797-804 (2024)

4. **P. Majumder**, S. Ahmed, P. Ahuja, A. Athreya, R. Ranjan, A. Penmatsa; Cryo-EM structure of antibacterial efflux transporter QacA from *Staphylococcus aureus* reveals a novel extracellular loop with allosteric role; The EMBO Journal, 42:e113418 (2023)

5. S. Dubey, **P. Majumder**, A. Penmatsa, A. A. Sardesai; Topological analyses of the L-lysine exporter LysO reveal a critical role for a conserved pair of intramembrane solvent-exposed acidic residues; Journal of Biological Chemistry, 297(4) 101168 (2021)

6. B. Khatri, **P. Majumder**, J. Nagesh, A. Penmatsa and J. Chatterjee; Increasing protein stability by engineering the n→π* interaction at the β-turn; Chemical Science, 11, 9480-9487 (2020)

7. **P. Majumder**, S. Khare, A. Athreya, N. Hussain, A. Gulati and A. Penmatsa; Dissection of protonation sites for antibacterial recognition and transport in QacA, a multi-drug efflux transporter; Journal of Molecular Biology, 431, 2163-2179 (2019)

8. **P. Majumder**, A. K. Mallela and A. Penmatsa; Transporters Through the Looking Glass: An Insight into the Mechanisms of Ion-Coupled Transport and Methods That Help Reveal Them; Journal of the Indian Institute of Science, 98, 283-300 (2018)

Scientific engagement

Invited and selected lectures

2027 | **Bose Institute, Kolkata**
Invited lecture | Upcoming; date to be announced

October 2026 | **EMSI 2026, Kolkata**
Invited lecture on Chp1 membrane pore activation and lipid gating

July 2026 | **Bhabha Atomic Research Centre, Mumbai**
Invited lecture on the Cap1 membrane pore complex and bacterial antiviral defence

May 2026 | **Symposium on the Immune System of Bacteria**
Selected speaker | Mechanistic diversity of cOA-activated CARF membrane effectors in CRISPR defence

November 2025 | **SciensPUR**
Invited speaker | Diverse immune systems in bacteria

July 2025 | **US Cryo-EM National Facilities Current Practices Webinar**
Invited speaker | Structural basis of a filamentous CRISPR effector revealed by cryo-EM

July 2025 | **EMSI 2025, Indian Institute of Science, Bengaluru**
Keynote speaker | CRISPR beyond nucleases

Selected research presentations and training

October 2026 Attended the G. N. Ramachandran Lecture Series on Bioinformatics and Computational Biology, IIT Kharagpur.

October 2025 CryoET Workshop, Brookhaven National Laboratory.

June 2025 Fifth Annual NY Area Cryo-EM Meeting, NYU Langone; oral presentation on Cat1 filament networks and NAD degradation.

October 2024 MSKCC Basic Science Retreat; oral presentation on Cat1-mediated antiviral defence.

June 2024 Fourth Annual NY Area Cryo-EM Meeting, Columbia University Medical Center; oral presentation on Cad1 and Cam1.

May 2024 MSKCC Structural Biology Retreat; oral presentation on Cad1.

October 2019 EMBO Workshop, EMBL Hamburg; poster on QacA substrate recognition and transport.

Professional memberships

Electron Microscope Society of India | Life member, 2026

Society of Biological Chemists (India) | Life member since August 2026

Research recognition and outreach

IIT Kharagpur Foundation feature | July 25, 2026
Research on the molecular machinery of bacterial immunity featured in the Foundation newsletter. Read feature

Research coverage | Rockefeller University press releases highlighted the Cat1 and Cad1 studies, with additional coverage of Cad1 in Genetic Engineering & Biotechnology News and a commentary in Cell Host & Microbe.